

Day – 6 Troubleshooting tools.

Task – 1 Use command to find ipv4 address.

- How to Find Your IPv4 Address and Default Gateway
- Press the Windows Key, type cmd, and press Enter to open the Command Prompt.
- Type the following command and press Enter:
- DOS
- ipconfig
- Look through the output for your active network connection (usually labeled as Wireless LAN adapter Wi-Fi or Ethernet adapter).
- Locate these two key lines:
- IPv4 Address: (e.g., 192.168.1.15) – This is your computer's unique address on your local network.
- Default Gateway: (e.g., 192.168.1.1) – This is the IP address of your home router.

Task – 2 Testing the connectivity.

-  Open your command-line tool:
- Windows: Press the Windows Key, type cmd, and press Enter to open the Command Prompt.
- Mac / Linux: Press Cmd + Space, type Terminal, and press Enter.
-  Type the ping command and press Enter:
- Bash
- ping www.google.com
-  Look at the results: After a few seconds, the command will send packets and display a summary at the bottom.

Task – 3 Run the tracert command.

-  Open your command-line interface:
- Windows: Open the Command Prompt (cmd).
- Mac / Linux: Open the Terminal app.
-  Type the traceroute command and press Enter:

- On Windows: `tracert [www.flipkart.com](https://www.flipkart.com)`
- On Mac/Linux: `traceroute [www.flipkart.com](https://www.flipkart.com)`
-  Count the hops: Watch as the list populates line by line. Each numbered line represents a router (hop) that your data packet traverses on its way to Flipkart's servers.

Task – 4 Finding the ip address of the application.

- **Running `nslookup [www.spotify.com](https://www.spotify.com)` queries the Domain Name System (DNS) to return its associated IP addresses (often routing through CDN endpoints like 35.186.224.24 or Fastly caching nodes such as 151.101.7.42).**
- **What it helps verify: The `nslookup` tool helps you verify whether domain name resolution is working correctly and which specific server IP addresses a domain is pointing to on the network.**

Task – 5 Based on scenario.

- 1. `ipconfig`
- What it helps check: It verifies the local network configuration on your friend's laptop. It ensures the device has a valid local IP address and is successfully communicating with the local router (default gateway). If this fails, the issue is local to the Wi-Fi or network adapter.
- 2. `nslookup [www.zomato.com](https://www.zomato.com)`
- What it helps check: It tests DNS resolution. This command queries whether the laptop can translate the domain name `[www.zomato.com](https://www.zomato.com)` into its corresponding server IP address. If this fails, it indicates a DNS configuration or server issue preventing the browser from finding where Zomato lives.
- 3. `ping [www.zomato.com](https://www.zomato.com)` (or `ping 8.8.8.8`)
- What it helps check: It checks end-to-end network connectivity and server reachability. By sending ICMP packets to Zomato's server (or a public DNS like Google if the domain fails), it measures whether data can successfully travel back and forth across the internet and detects any packet loss.
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